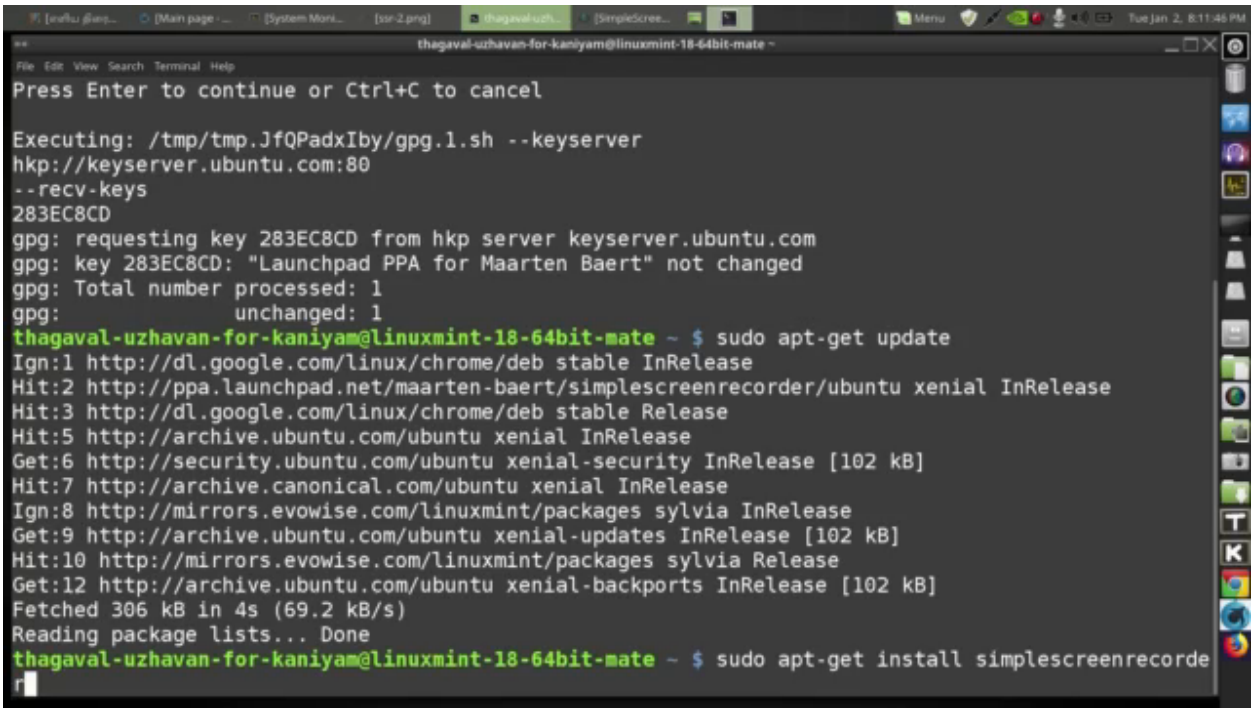
Frame Review: Keep the Lesson

Keep the command. Calm the blink.
Joseph Ayanda / AI-assisted development / 23 October 2026



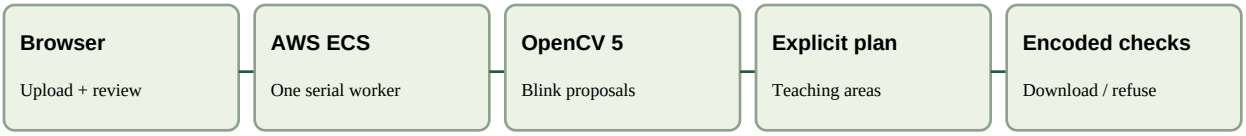
Real installation tutorial by Info-farmer (2018), CC0 1.0, via Wikimedia Commons. Extracted seconds 68-80, resized and re-encoded. No visual defect was added. This is original creator narration, not Joseph's final demo voice.

The problem. A lesson editor has a flattened recording, with no editable project. A caret keeps blinking after the command has been typed. Changing the whole video would change the lesson. Frame Review offers a smaller decision: keep one observed caret state during a stationary pause, while preserving the command, timeline and prepared narration.

Verified result on this case	What it means
6 changes -> 0 in frames 30-149	One natural four-second interval inside a twelve-second excerpt. The count excludes entry and return.
360 protected RGB frames exact	Command area [0,0,640,336] matches the prepared input on every decoded frame.
Prepared audio exact	AAC packets, timing, skip metadata and decoded PCM are unchanged by the edit.

Current evidence: the untouched 34,393,948-byte original completed upload, preparation, review, explicit approval, export and downloads through the public AWS HTTPS app. Its output is byte-identical to the independently decoded local qualification. This is one targeted natural lesson case, not a measured success rate or a teacher trial.

Vision proposes. The editor decides.



HTTPS load balancer / digest-pinned ECR image / Fargate CPU / CloudWatch logs

FFmpeg preparation and export run inside ECS. Review never approves itself.

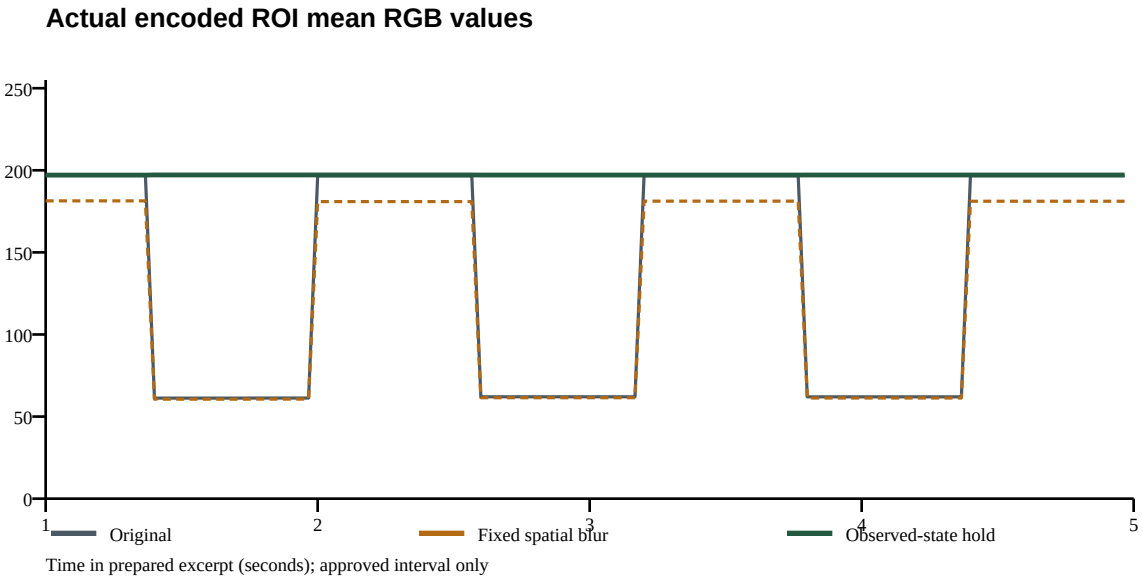
Substantive OpenCV 5. The new steady-detail-v1 detector measures native luma state changes, connected components and small stationary support. It checks two observed states, minimum dwell and bounded luma/chroma variation. OpenCV also creates overlays and exact encoded RGB crops, accumulates error histograms and measures declared teaching areas. NumPy holds native YUV planes and mappings. FFmpeg performs bounded decoding, preparation, lossless encoding and audio transport.

A separate versioned action. The earlier rapid-change detector and 5x periodic retimer remain frozen. They missed this ordinary caret. The new action holds the observed state with greater contrast against its surrounding ring, only in its proposed box and interval. It does not invent pixels or recognize teaching meaning. A source-changing instruction, counter or cursor can still be useful: the editor must compare both states and reject an unsuitable proposal.

The preservation contract. A steady-detail edit needs at least one named teaching rectangle. Overlap with the edit's four-pixel RGB influence area is refused. Approval binds the source, both analyses, exact proposal, implementation and teaching areas. The worker recomputes the proposal before exporting. Native samples outside [10,340,8,14], all 240 frames outside the interval, all RGB outside the influence area, every declared teaching frame and supported prepared audio must pass their checks. Rejected bytes never appear as an accepted edited download.

Preparation is a real limitation. MP4/MOV or WebM intake extracts 1-12 seconds, letterboxes to 640x360, converts to 30 fps and encodes H.264 and optional mono 48 kHz AAC once. This changes original picture/audio and can erase small labels. Exact edit checks reference that prepared clip. The original upload and prepared source remain independently hash-bound and downloadable. Preserved pixels do not prove readable teaching.

Measured change, with a reference



Method on the same interval	Adjacent ROI-mean changes >=8 RGB codes
Original prepared lesson	6
One fixed Gaussian spatial blur	6
Observed-state hold	0

The declared blur reference uses a 5x5 Gaussian kernel, sigma 2, on native luma inside the same box and interval, retaining source chroma. Its actual encoded output was decoded and measured. It preserved the outside area and prepared audio but left the repeated activity. This is one simple, untuned reference, not a comparison with a complete professional editor or a user study.

What stayed visible. The new action keeps a state actually observed in the recording, including its native caret samples. The commands stay on their own timeline. Independent local decoders checked native outside samples, intended mapping, all-frame teaching RGB, outside influence RGB and prepared PCM. The cloud output has the same SHA-256, so it has the same encoded bytes. All twelve browser-downloaded boundary crops also matched their report hashes.

What this does not show. The six-to-zero count covers the interval interior. Entry and return are separately reported: source return delta 0.61055, edited return 0.61449 in the existing selected-luminance diagnostic. The return remains strong. No attention, comprehension, accessibility, medical, time-saving or adoption benefit has been measured.

Negative evidence remains. The earlier six-source, fourteen-excerpt HTTP audit found no qualified natural repairs with the rapid-change engine. A potentially instructional Blender field was protected and refused; two separate camera-flicker exports failed recurrence. Four additional licensed tutorials were screened on fixed grids in this development search and produced no supported stationary proposal. Two tiny browser-caret hypotheses also failed after actual preparation. These are convenience/targeted observations, not a representative recall estimate. The added-corner classroom challenge remains a disclosed controlled positive.

Reproduce the qualified release

Live AWS execution verified. https://le-d72c965de52749b08fd72a2513bccce1.ecs.us-east-1.on.aws
One completed ECS/Fargate revision, one running task. Actual original upload -> preparation -> OpenCV review -> approved export -> original/prepared/output/report/decision downloads. Reload recovery, 375/890 layouts, twelve crop hashes and unowned-job 404s passed. The downloaded file played through twelve seconds at 640x360 without a media error in a separate browser player; listening was not evaluated.

Qualification. All eleven verification stages passed in the exact image: 137 backend tests, numerical/duration/audio/pixel refusal suites, authored lesson, real classroom intake and disclosed challenge, natural steady-detail export and 73 studio tests (72 passed, one platform skip). Source identity is separate from these measured media results.

Fast judge route. Open the live app, or build the Dockerfile and run `python studio/run.py --serve --port 8877` inside the Linux image. Click **Try a real terminal blink**. Compare both source states. Name and add teaching area **[0,0,640,336]**, check the plan, then explicitly generate a comparison. Inspect all twelve crops, the six-to-zero activity result and protected/audio measurements. Download and review the edited clip before use. The included preset is a disclosed prepared excerpt of the same natural case, not another independent success.

Operation and limits. Sources up to 64 MiB and 30 minutes; one supported 8-bit SDR video up to 3840x2160 and at most one supported mono/stereo audio track. Excerpt start 0-600 seconds, length 1-12 seconds. No full-lesson splice-back. The worker runs as UID 10001 with process/time/memory/disk caps, file/pipe protocols, one queue and eight job slots. Secure/HttpOnly/SameSite cookies and host/origin/request-token checks scope access. No application AWS API role is needed. Jobs expire after 24 hours and a restart can clear them sooner. Known package findings remain; restrictions are mitigations, not certification.

Competition fit. Education is the concrete use case. Technical depth is in substantive OpenCV and encoded preservation. Innovation is a reviewer-controlled, source-bound local edit. Impact evidence is a natural case plus a measured reference; broad benefit remains unmeasured. UX includes both observed states, teaching protection, refusal recovery and reload. AWS performs the media work. No COOL or Agentic Vision prize claim is made. Joseph's real-voice final demo and creator/user trials remain outstanding.

Runtime source files: 96
Manifest SHA-256: 593b07a52cee2dda00fd756e387f9059f9a7c6ee7c5026cebf42c8f6fb86b658
ECR image digest: sha256:0991b45074e726f43492ed480222ec3943888f1b13a8534d1e8b5dd1a468af15
Natural output SHA-256: ce027d46d71a225729a56f25c4c68cf78bdd9f60769db7da26a5a01991e2e0b6
Code, report and receipts: https://github.com/josepha-mayo/opencv-frame-review
Source: https://commons.wikimedia.org/wiki/File:Tutorial-tamil-installation-SimpleScreenRecorder.webm
License: https://creativecommons.org/publicdomain/zero/1.0/